

PRACTICAL NO 1

Aim: Write a program to implement CRUD operations.

INPUT:

```
class Student:
    next_id = 1
    def __init__(self, name, elective):
        self.studentId = Student.next_id
        self.studentName = name
        self.selectedElective = elective
        Student.next_id += 1
    student_db = []
    def add_students(self):
        Student.student_db.append(self)
        print(f"{self.studentId} | {self.studentName} | {self.selectedElective} is Added in database")

    def update_elective(self, new_elective):
        for student in Student.student_db:
            if student.studentId == self.studentId:
                student.selectedElective = new_elective
                return "Student updated successfully"
        return "Student not found"

    def remove_student(self):
        for i in range(len(Student.student_db)):
            if Student.student_db[i].studentId == self.studentId:
                print(f"\n\nRemoving :- {self.studentName}")
                Student.student_db.pop(i)
                return "Student removed successfully"
        @classmethod
        def display_database(cls):
            print("\n\n Current student records ")
            for student in Student.student_db:
                print(f"ID: {student.studentId} | Name: {student.studentName} | Elective : {student.selectedElective} ")

Nahid = Student("Nahid", "IOT")
Nahid.add_students()
yusuf = Student("Yusuf", "ML")
yusuf.add_students()
ismail = Student("Ismail", "IOT")
ismail.add_students()
fareed = Student("Fareed", "AI")
fareed.add_students()

Student.display_database()
Nahid.update_elective("AI")
Student.display_database()

print(Nahid.remove_student())
Student.display_database()
```

OUTPUT:

```
1 | Nahid | IOT is Added in database
2 | Yusuf | ML is Added in database
3 | Ismail | IOT is Added in database
4 | Fareed | AI is Added in database
```

```
Current student records
ID: 1 | Name: Nahid | Elective : IOT
ID: 2 | Name: Yusuf | Elective : ML
ID: 3 | Name: Ismail | Elective : IOT
ID: 4 | Name: Fareed | Elective : AI
```